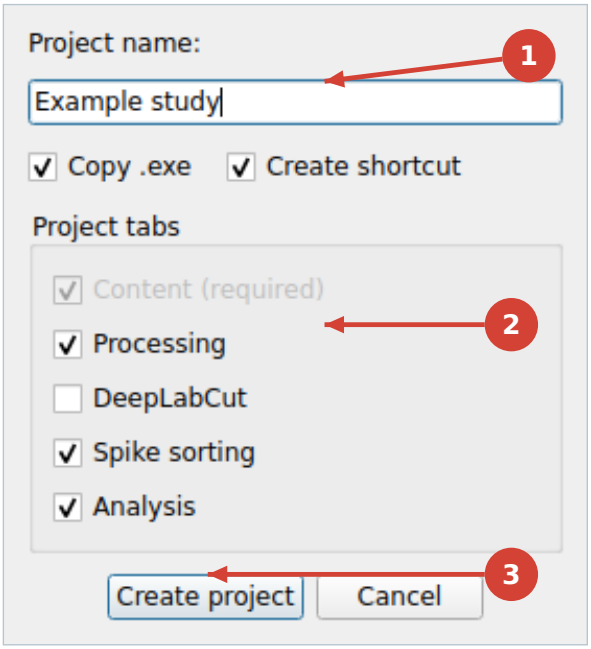
1 Create your project

Start with converted NWB files. The project keeps your file list, settings, scripts and results together.



1. From the launcher, click **Create project**. Choose a working folder and enter a project name (1).

2. Enable the tabs you need (2). DeepLabCut and Spike sorting are optional.

3. Click **Create project** (3). Reopen it later from the launcher, its shortcut or its project launcher script.

The workflow in one sentence

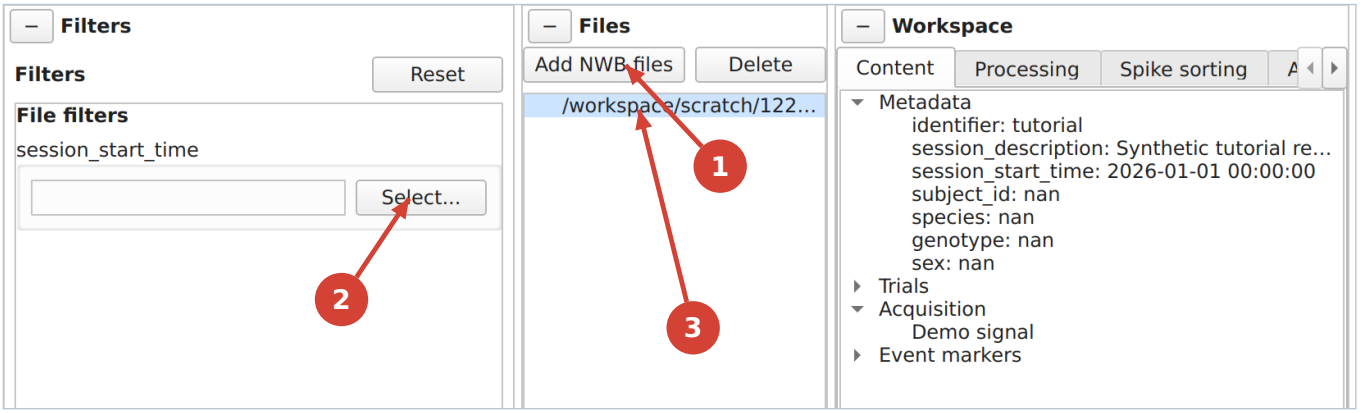
Add recordings, select files, inspect their contents, create any needed derived data, then analyze and save results.

Creating a project does not copy or merge your NWB recordings. Keep source NWB files and externally linked videos accessible.

This short guide uses real application screenshots with illustrative data. Some general-view captures are from an earlier build; the same main controls apply. Detailed settings and scripting are in the complete guide.

2 Add, filter and select files

Filters decide what is visible. Selection decides which files you work on.



1: Add NWB files. 2: choose metadata values. 3: select a recording or folder in Files.

1. Click **Add NWB files**. Select NWB files or a folder to scan. Wait for indexing to finish.

2. Use the left **Filters** panel to narrow the list, for example by subject, session or protocol. Available fields come from your NWB metadata.

3. Click files or folders in **Files**. Use Ctrl/Shift for a subset, or **Select all** for the current visible list. **Clear selection** deselects it.

Remember these three rules

Values within one filter are alternatives: session A **or** B. Different filters combine: subject X **and** session A.

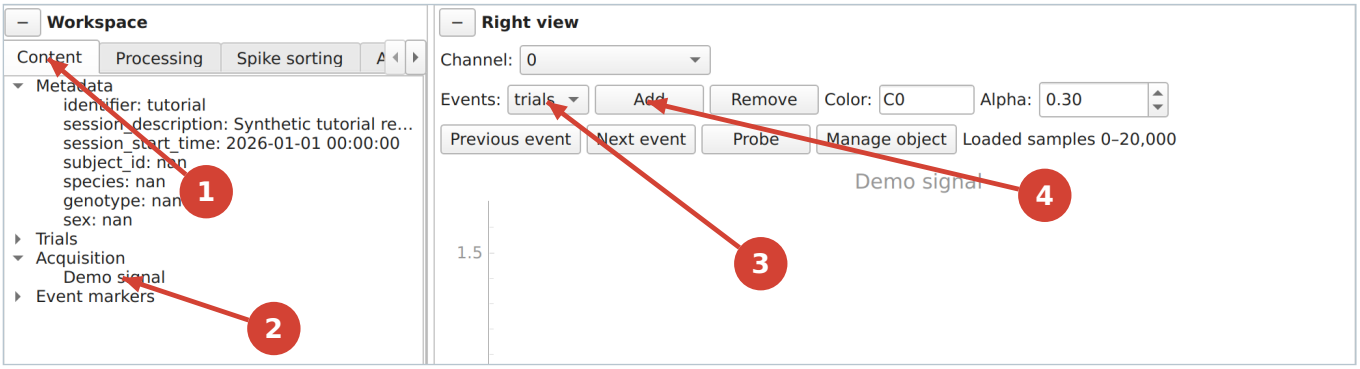
After changing a file filter, check the selection again. A visible file is not necessarily selected.

Delete in Files removes entries from the project list. It keeps the NWB files on disk.

Check the Content summary after adding data. It tells you which objects are present and how many files contain them.

3 Inspect data and events

Use Content to check the signal, video and timing before running a batch.



1: Content tab. 2: open an object. 3: choose an event. 4: Add its overlay.

1. Select a recording in Files, open **Content** (1), then expand **Acquisition**, **Processing**, **Units** or **Trials** and click an object (2).

2. For signals, choose a channel and time window. Left-drag to move in time. For video, use Play, frame stepping and replay speed; wheel-zoom and drag to inspect details.

3. Choose an **event marker** (3), click **Add** (4), then use Previous/Next event to inspect timing.

What the objects mean

Acquisition: recorded signals/videos. **Processing:** derived objects. **Units:** spike times. **Trials:** intervals with trial metadata.

An **event marker** says when something happened. Overlaying it shows timing; it does not filter your trials. Analysis alignment uses these markers to cut out time windows.

Linked views share NWB time, including two signals or a signal plus video. This does not repair incorrectly synchronized timestamps.

4 Create a processed object

Use Processing when you want a derived signal or object saved inside the NWB file.



Example tool from the complete guide: input selector, output name and Run. Other tools have their own parameters.

1. Select the source files, open **Processing**, and choose a tool. Check **Available files** and read its description.

2. Choose the input object (1), enter the parameters and give the output a clear name (2).

3. Choose whether to replace an existing output with the same name. Click **Run** (3). Test one file first.

4. Return to **Content > Processing** and inspect the new object. Check its values, units, duration and timing.

Keep the distinction clear

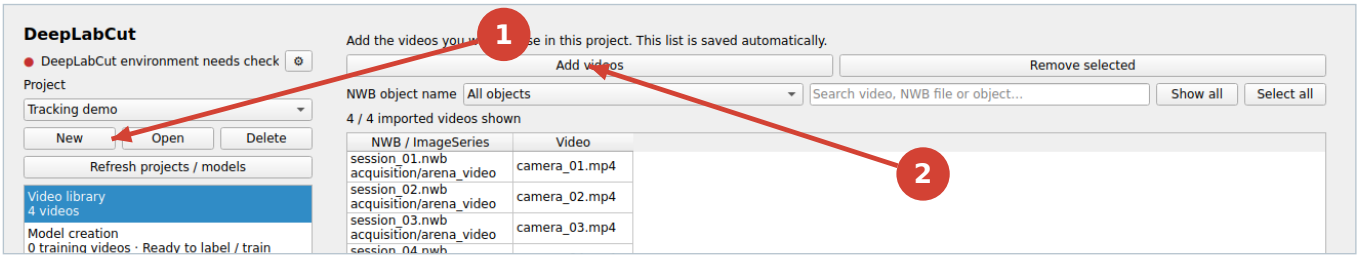
Processing usually writes derived data into NWB. **Analysis** usually saves figures and numerical summaries in the project folder.

Processing uses the selected files. It is not automatically limited to the current displayed time window or to Analysis trial filters.

For a repeated sequence, configure each step and use **Add to pipeline**. Check the saved order and settings before launching.

5 DeepLabCut: label and train

Optional. Import videos once, then choose representative videos for model training.



1: create/open a DLC project. 2: add video references to its saved library.



3: Open labeling. 4: Train model. Training parameters and Evaluate model are alongside them.

1. Check the detected DLC environment, then click **New** or **Open** (1).

2. In **Video library**, add videos from the selected NWB files (2). Filter by NWB object name to choose the camera. The library persists when you close the app.

3. In **Model creation**, choose/create a named model and select training videos. Use a representative subset; random selection is a starting point.

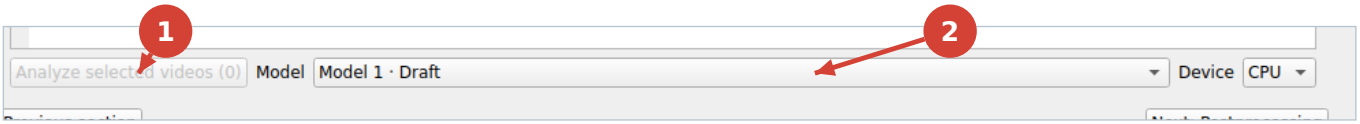
4. Open **Parameters** to set body parts and frames per video. Click **Open labeling**: frames are extracted when needed and Napari opens. Label consistently and save.

5. Open **Training parameters**, check device/epochs, then **Train model**. The training dataset is prepared automatically. Use **Evaluate model** after training.

A trained model locks its training selection. Create another named model to train a different selection. Labeling saves annotations; training learns from them.

6 DeepLabCut: analyze and refine

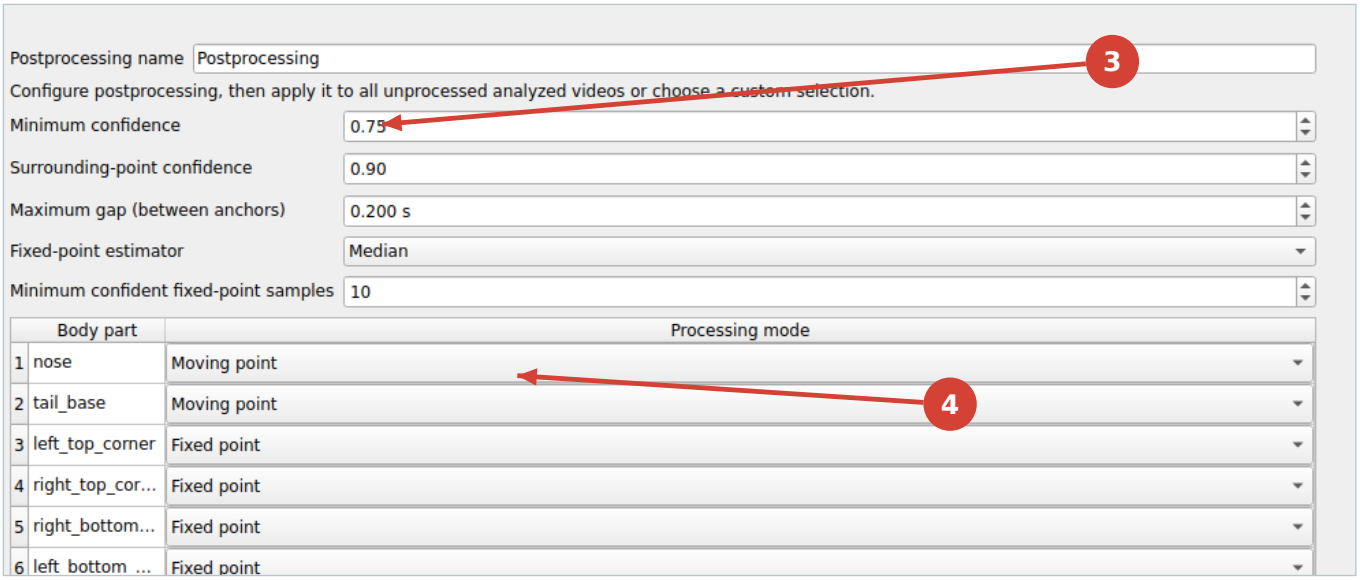
Apply a trained model to selected videos. Postprocessing is a separate, optional action.



Choose the trained model and device beside Analyze selected videos. GPU is preferred when detected.

1. Open **Analyze videos**, select the model and check the videos to process. **Select not analyzed** helps continue a batch.

2. Click **Analyze selected videos** (1). The table updates after each completed video. Check the log for failed or partial results.



Postprocessing: set confidence/gap rules, then choose the mode separately for each body part.

For moving points, use **Moving point** to interpolate short low-confidence gaps between confident detections. For stationary landmarks, use **Fixed point** to estimate one mean/median position for the video.

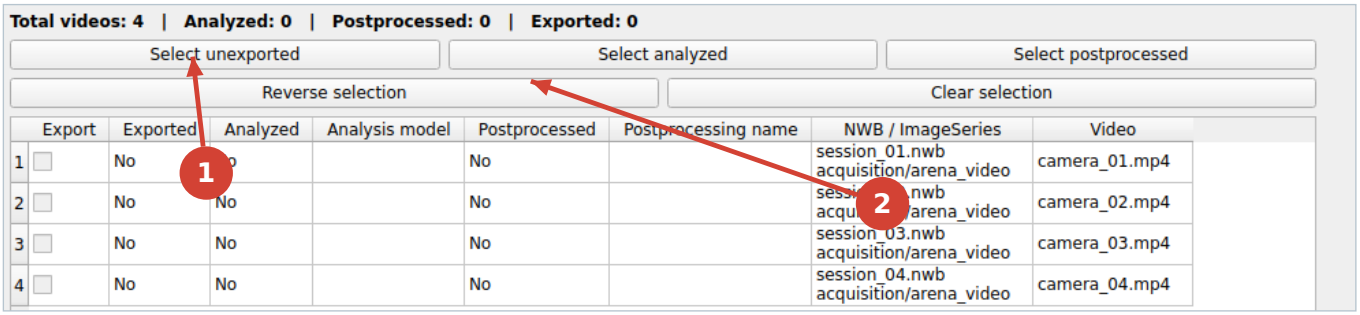
Name the operation, then **Apply to all unprocessed videos**, or open **Custom video selection** and highlight a subset. Changing parameters alone does not apply them.

Completed video predictions are saved during the batch. Canceling should retain completed videos; the interrupted video is not a finished result.

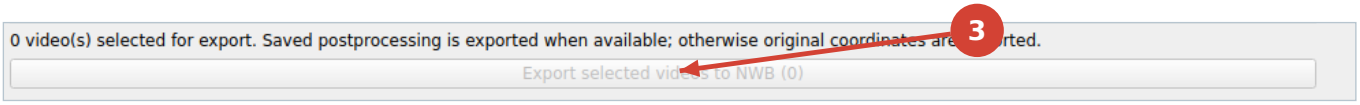
7 DeepLabCut: review and export

Review predictions first. Export makes tracking available in the NWB Content view.

In **Review**, select a video to load it. Zoom, step frames and compare **Processed** with original positions. Adjust dot size and confidence for readability. Filled circles are original points, hollow circles interpolated points, and squares fixed estimates.



Export has its own selection and shows analysis, model, postprocessing and export status.



The export button includes the number of selected videos.

1. Open **Export**. Use **Select unexported**, **Select analyzed** or **Select postprocessed**, then adjust the checked videos.

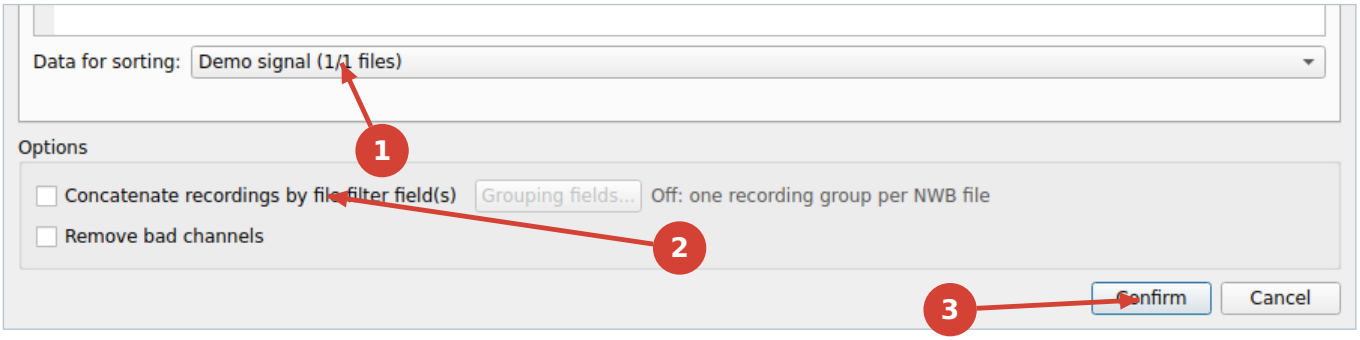
2. Click **Export selected videos to NWB** (3). Read the summary and choose whether to skip or replace existing corresponding tracking objects.

3. Watch the per-video status/log. Reopen a source file in **Content > Processing** to inspect the exported body parts.

Export uses saved postprocessing when available, otherwise original predictions. It does not run new postprocessing. Each body-part object contains X/Y, original confidence and processing status.

8 Spike sorting

Optional. Use recordings that retain spike-band information, normally raw extracellular data.



New sorting dialog, illustrated with demo data only. Choose suitable raw electrophysiology, not the tutorial signal.

1. Select recordings in Files, open **Spike sorting > New**, enter a name and choose **Data for sorting** (1).

2. Leave concatenation off for separate recordings. To join compatible recordings, enable it (2), choose grouping fields and check file order/channels. Click **Confirm** (3).

3. In **Preprocessing**, check channels, probe, filters and reference. Inspect representative traces.

4. In **Sorting**, choose the sorter and parameters, then **Add configuration** and **Run sorting queue**. Use **Unit summary parameters** to set event-aligned figures.

5. Inspect unit summaries. In **Curation**, open the editor, review units, save decisions, then apply the saved curation in Workbench.

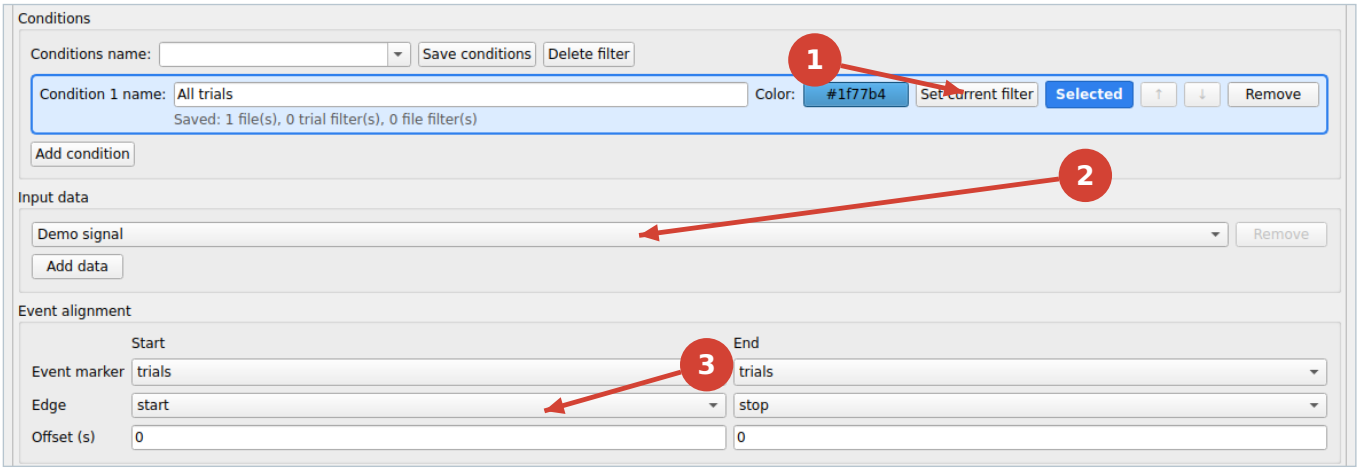
6. Use **Export**, select the intended run/curated variant, and choose **Create Units in source NWB files**. Check the label/replacement options.

Concatenation joins samples for sorting without merging the source NWB files. Export maps spike times back to each source file. Verify the exported Units in Content.

Waveforms, correlograms and rasters help assess quality. A completed sorter run is not a substitute for manual quality review.

9 Analysis: define the comparison

Choose a method, capture conditions, then select its input data and event window.



1: Set current filter saves a condition. 2: input data. 3: Start/End alignment.

1. Open **Analysis** and choose a method. Read what it expects and what it calculates.

2. Set file filters and select files. Set trial filters if needed; these appear in Analysis when trial fields are available.

3. Name the condition and click **Set current filter** (1). Repeat for other conditions, then **Save conditions**.

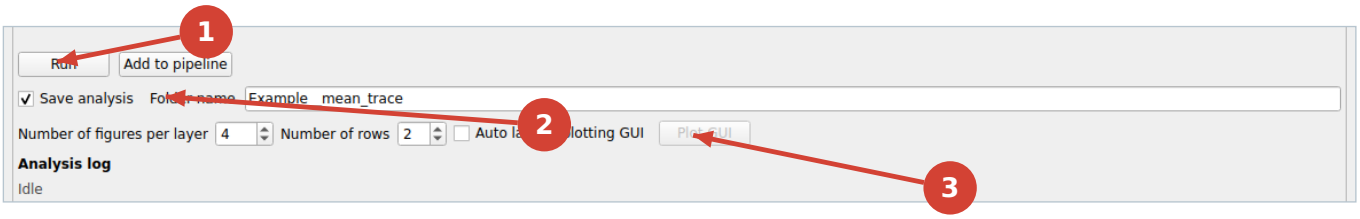
4. Select the **Input data** (2). Use **Add data** if the method accepts several objects. Some methods read NWB directly and need no input selection.

5. Set **Start**, **End**, edges and offsets (3). For complete trials, use trials/start/0 to trials/stop/0. Enter the method parameters.

A file filter selects recordings. A trial filter selects observations within them. Alignment selects the time window. Saved conditions keep their captured selection until you update them.

10 Run, inspect and save

Keep the figures, numerical outputs and log together so you can understand the result later.



1: Run. 2: Save analysis and folder name. 3: Plot GUI / automatic plotting.

1. Tick **Save analysis** and choose a descriptive folder name (2). Check conditions, data, timing and parameters one last time.

2. Click **Run** (1). Read the log for accepted files/trials, exclusions and errors.

3. Open **Plot GUI** (3), or enable automatic launch. Use Previous/Next, layout and axis controls to inspect the registered figures.

4. Find saved results under the project's **Analysis** folder, unless a different output path was chosen. Reopen a figure, its data table and **analysis_log.txt**.

Before running hundreds of files

Start with two or three representative recordings. Verify the timing in Content, confirm the selected files/trials, and check that the numerical output matches the figure.

Where each action leaves its result

Processing: derived objects in NWB.
DLC / sorting: project results first, then explicit NWB export.
Analysis: figures, data and logs in result folders.

If nothing is produced, check the first specific log message: missing object, empty selection, no valid trials, or failed quality threshold. Use the complete guide for advanced settings and script development.